

Cut Locus of Submanifolds: Stability and Intersection with Focal Locus

Aritra Bhowmick

Department of Mathematics and Statistics, School of Science

GITAM, Hyderabad

abhowmic@gitam.edu

Presented at

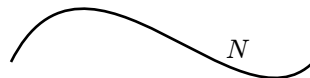
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$$p := \gamma(0) \in N, \quad \gamma(L) = q, \quad \text{Len}(\gamma) = L = \text{dist}_g(N, q).$$



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The **cut locus** $\text{Cut}(N)$ of a submanifold $N \subset M$ is the set of points $q \in M$ such that there exists an N -segment joining N to q , any extension of which fails to be distance-minimizing from N .

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Theorem : ([, Prasad [2024]][†])

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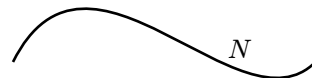
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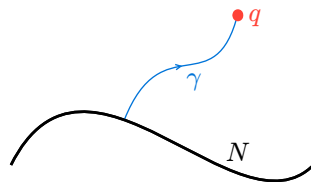
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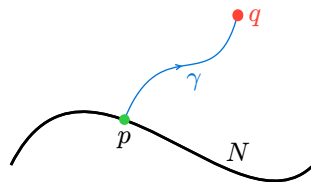
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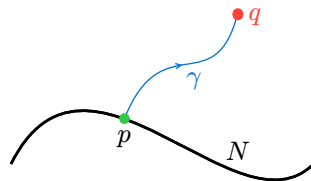
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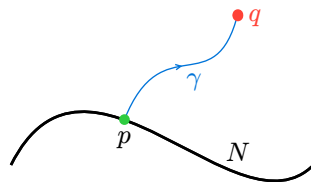
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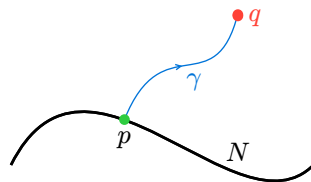
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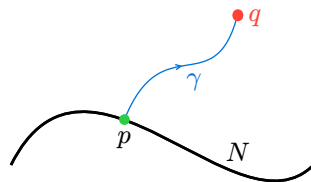
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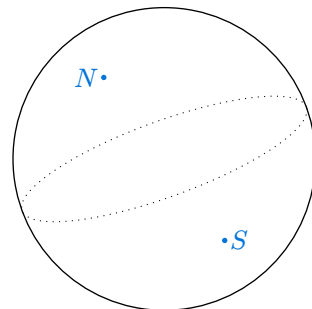
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Examples

The cut locus of the south pole S on the sphere S^2 consists of the north pole N .

More generally, the cut locus of a point $p \in S^n$ is the singleton $\{-p\}$



The cut locus of the equatorial S^1 in the sphere S^2 is the set $\{N, S\}$, which is S^0 .

More generally, the cut locus of $S^k \subset S^n$ is S^{n-k-1}

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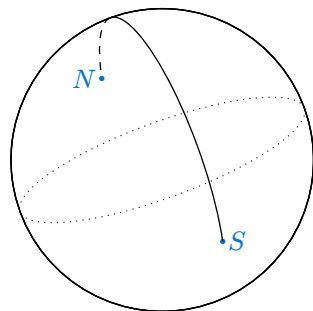
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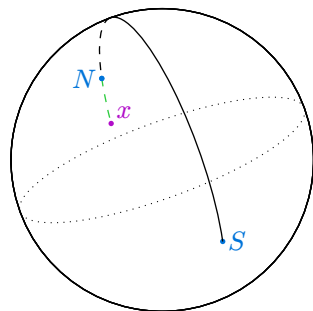
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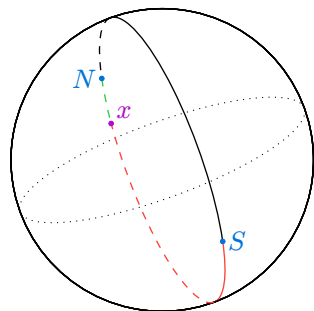
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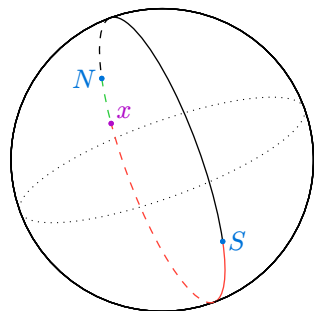
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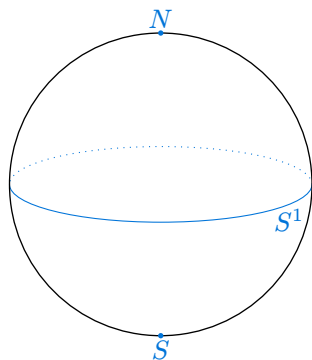
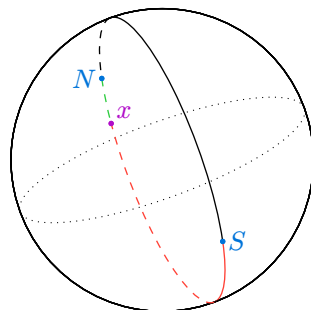
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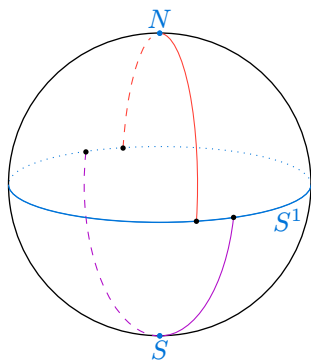
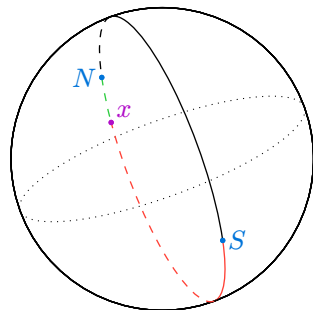
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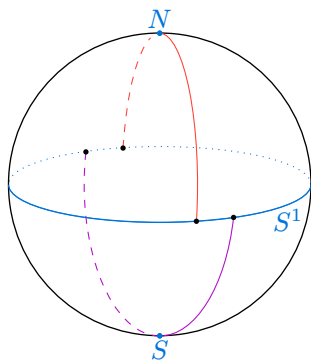
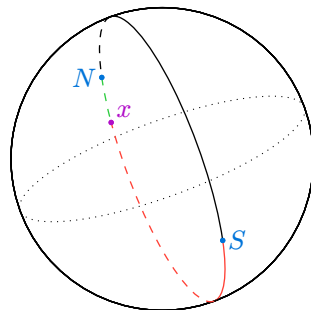
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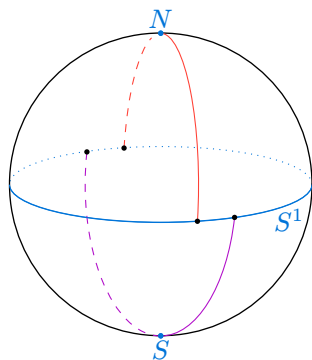
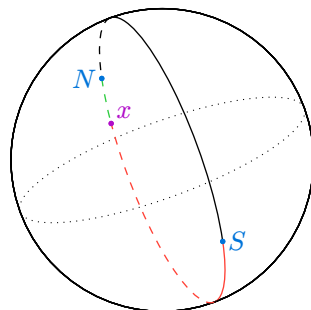
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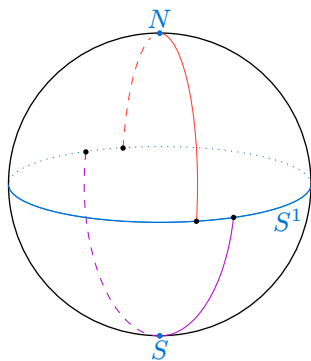
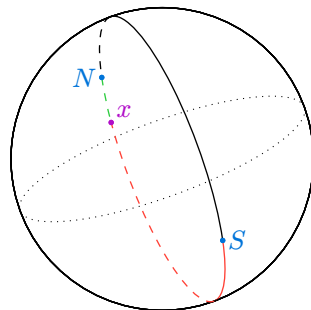
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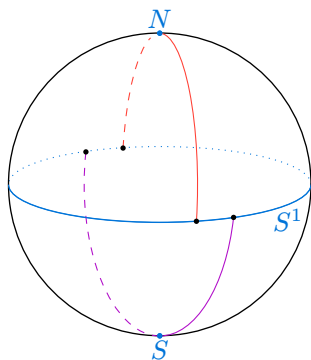
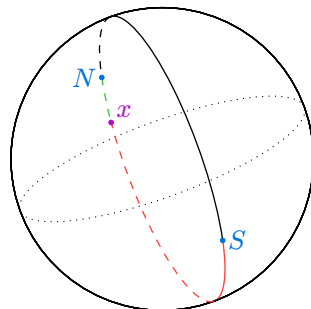
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The cut locus of $O(n, \mathbb{R})$ in $M(n, \mathbb{R})$ is $M(n, \mathbb{R}) \setminus \text{GL}(n, \mathbb{R})$, the set of singular matrices (Basu-Prasad [2023][†]). This gives a geometric interpretation of the Gram-Schmidt process.

[†] Somnath Basu and Sachchidanand Prasad. *A connection between cut locus, Thom space and Morse-Bott functions*. *Algebr. Geom. Topol.* (2023).

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Remark

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The stability of the cut locus is well-studied in the literature from different perspectives.

- **Buchner (1977)[†]**: $\text{Cut}(p, g)$ is stable from the point of view of singularity theory for generic g , on a manifold of dimension ≤ 6 .
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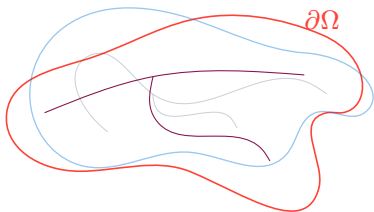
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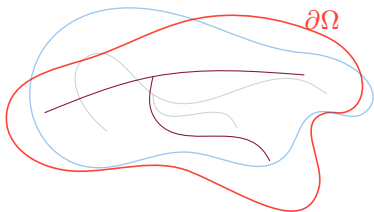
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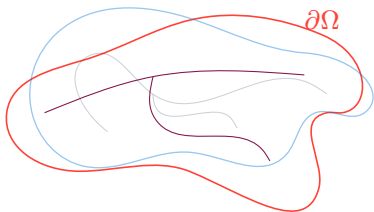
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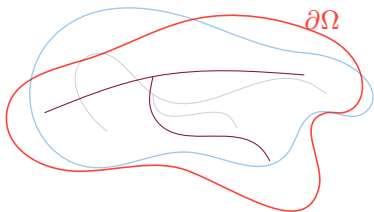
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Cut Locus as a Map

Given closed sets $A, B \subset M$ of a metric space (M, d) , the **Hausdorff distance** between them is

$$d_H(A, B) := \max \left\{ \sup_{a \in A} \text{dist}(a, B), \sup_{b \in B} \text{dist}(A, b) \right\}.$$

Let M be a compact manifold without boundary, and $\mathfrak{h}$ be an auxiliary metric on M .

- Let $\mathcal{G} = \mathcal{G}(M)$ be the collection of C^2 Riemannian metrics on M , equipped with the Whitney C^2 -topology.
- Let $\mathcal{K} = \mathcal{K}(M) := \{K \subset M \mid K \text{ is compact}\}$, equipped with the topology induced by the Hausdorff metric d_H associated with $\mathfrak{h}$. The topology is independent of the choice of $\mathfrak{h}$.
- Given a manifold N , let $\text{Emb}(N, M)$ be the space of smooth embeddings $N \hookrightarrow M$, equipped with the Whitney C^2 -topology.

As the cut locus is closed, we have the maps

$$\begin{aligned} \text{Cut} : M \times \mathcal{G} &\rightarrow \mathcal{K} \\ (p, \mathfrak{g}) &\mapsto \text{Cut}(p, \mathfrak{g}), \end{aligned}$$

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Theorem : ([Itoh, Prasad 2026][†])

The cut locus map $\text{Cut} : M \times \mathcal{G} \rightarrow \mathcal{K}$ is continuous.

Recently, we have generalized the result of Sakai [1983].

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Proof Sketch of Stability: Ingredients

We have the characterization of convergence in the Hausdorff metric.

Lemma

Suppose M is compact, $X_i, X \subset M$ are closed (and hence compact) sets. Then, $\lim_i d_H(X_i, X) = 0$ if and only if the following holds:

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The distance function $u := \text{dist}_g(\partial\Omega, \cdot)$ is *not* C^1 on points of $\text{Sep}^{\text{in}}(\Omega, g)$, and the C^1 singular support of u inside Ω is precisely $\text{Cut}^{\text{in}}(\Omega, g)$.

Lemma

$x \in \text{Sep}^{\text{in}}(\Omega, g)$ if and only if there exists a C^1 function φ near x such that $u - \varphi$ attains a (strict) local maximum at x and $\|d_x \varphi\|_g < 1$.

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We have the characterization of convergence in the Hausdorff metric.

Lemma

Suppose M is compact, $X_i, X \subset M$ are closed (and hence compact) sets. Then, $\lim_i d_H(X_i, X) = 0$ if and only if the following holds:

- for any $x \in X$, there exists $x_i \in X_i$ with $\lim_i x_i = x$, and
- for any convergent sequence $x_{i_j} \in X_{i_j}$, we have $\lim_j x_{i_j} \in X$.

We also have a useful characterization of a separating point. Given $\Omega \subset M$ open, consider the cut locus (resp. separating set) of the boundary $\partial\Omega$ inside Ω :

$$\text{Cut}^{\text{in}}(\Omega, \mathbf{g}) := \text{Cut}(\partial\Omega, \mathbf{g}) \cap \Omega, \quad \text{Sep}^{\text{in}}(\Omega, \mathbf{g}) := \text{Sep}(\partial\Omega, \mathbf{g}) \cap \Omega.$$

The distance function $u := \text{dist}_{\mathbf{g}}(\partial\Omega, \cdot)$ is *not* C^1 on points of $\text{Sep}^{\text{in}}(\Omega, \mathbf{g})$, and the C^1 singular support of u inside Ω is precisely $\text{Cut}^{\text{in}}(\Omega, \mathbf{g})$.

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$x \in \text{Sep}^{\text{in}}(\Omega, \mathbf{g})$ if and only if there exists a C^1 function φ near x such that $u - \varphi$ attains a (strict) local maximum at x and $\|d_x \varphi\|_{\mathbf{g}} < 1$.

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Lemma

Suppose $g_i \rightarrow g_0$ is a C^2 -convergent sequence of metrics. Given $x \in \text{Cut}(N, g_0)$, there exists a sequence $x_i \in \text{Cut}(N, g_i)$ such that $\lim_i x_i = x$.

Proof : Since $\text{Cut}(N, g_0) = \overline{\text{Sep}(N, g_0)}$, assume $x \in \text{Sep}(N, g_0)$. As $\lim_i \text{Inj}(N, g_i) = \text{Inj}(N, g_0) > 0$, we have $\delta > 0$ such that $\delta < \text{Inj}(N, g_i)$. Fix the sets and the distance functions:

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Proof of the Continuity : The proof of the continuity of the cut locus map now follows from the previous two lemmas, and the characterization of the convergence in the Hausdorff metric. □

Corollary

Suppose $g_i \rightarrow g_0$ is a C^2 -convergent sequence of metrics, and $f_i \rightarrow f_0$ is a C^2 -convergent sequence of embeddings $N \hookrightarrow M$; set $N_i = f_i(N)$. Let $\mathbf{n}_i \in TN_i^{\perp_{g_i}}$ be unit normal vectors converging to some $\mathbf{n}_0$, necessarily in $TN_0^{\perp_{g_0}}$ with $\|\mathbf{n}_0\|_{g_0} = 1$. Then, $\lim_i \rho_{g_i}(N_i, \mathbf{n}_i) = \rho_{g_0}(N_0, \mathbf{n}_0)$.

The above strengthens the classical result about the continuity of the cut time map of a fixed submanifold, with respect to a fixed metric.

Proof Sketch of Stability (cont.)

Lemma

Suppose $\mathbf{g}_i \rightarrow \mathbf{g}_0$ is a C^2 -convergent sequence of metrics. Given $x_{i_j} \in \text{Cut}(N, \mathbf{g}_{i_j})$ with $x = \lim_j x_{i_j}$, we have $x \in \text{Cut}(N, \mathbf{g}_0)$.

Proof of the Continuity : The proof of the continuity of the cut locus map now follows from the previous two lemmas, and the characterization of the convergence in the Hausdorff metric. $\square$

Corollary

Suppose $\mathbf{g}_i \rightarrow \mathbf{g}_0$ is a C^2 -convergent sequence of metrics, and $f_i \rightarrow f_0$ is a C^2 -convergent sequence of embeddings $N \hookrightarrow M$; set $N_i = f_i(N)$. Let $\mathbf{n}_i \in TN_i^{\perp_{\mathbf{g}_i}}$ be unit normal vectors converging to some $\mathbf{n}_0$, necessarily in $TN_0^{\perp_{\mathbf{g}_0}}$ with $\|\mathbf{n}_0\|_{\mathbf{g}_0} = 1$. Then, $\lim_i \rho_{\mathbf{g}_i}(N_i, \mathbf{n}_i) = \rho_{\mathbf{g}_0}(N_0, \mathbf{n}_0)$.

The above strengthens the classical result about the continuity of the cut time map of a fixed submanifold, with respect to a fixed metric.

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Focal Points

A cut point may not always be a separating point.

Let N be an ellipse in $\mathbb{R}^2$, with focal points e, e' .

$\text{Sep}(N) = \text{Open interval between } e \text{ and } e'.$

For some x on the open interval, we have $[p_1, x]$ and $[p_2, x]$ as N -segments. For e (resp. e'), we have *unique* N -segment $[a, e]$ (resp. $[b, e']$), any extension of the of which fails to be distance minimizing from N . Thus,

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Definition

Given $N \hookrightarrow M$, a point $q \in M$ is called a **focal point** of N if it is a *critical value* of the normal exponential map $\exp^\perp : TN^\perp \rightarrow M$

Theorem : ($\square$, Prasad [2024])[†]

For any $q \in \text{Cut}(N)$, we have a *non-exclusive* dichotomy:

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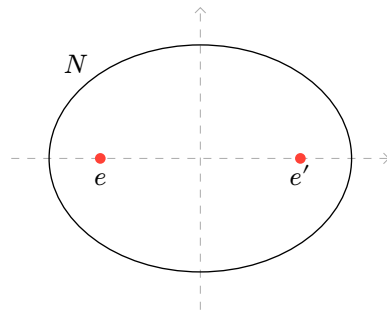
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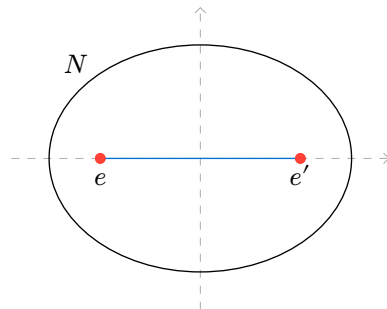
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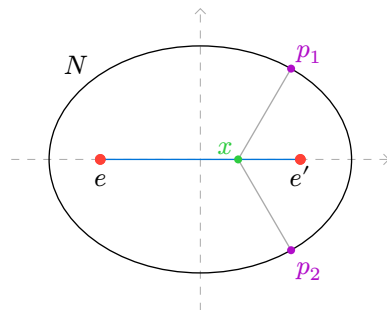
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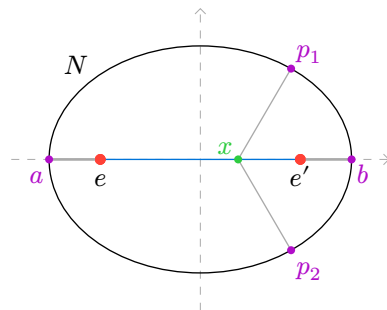
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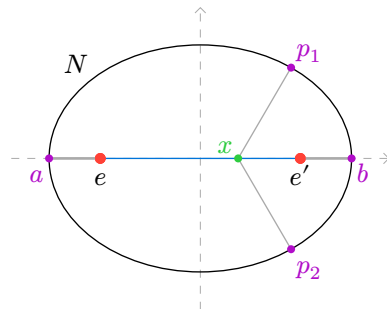
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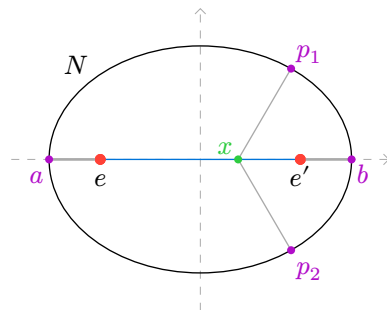
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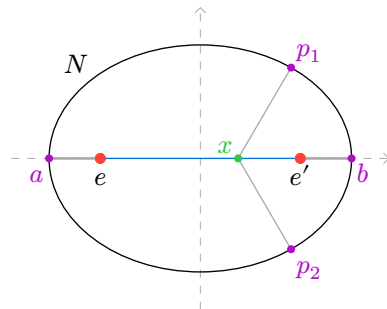
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Intersection of Cut and Focal Loci

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For any point p in a compact, simply connected, Riemannian manifold (M, g) , the cut locus $\text{Cut}(p, g)$ must have a conjugate point of p along some distance-minimizing geodesic from p .

The conjecture is true for S^2 , and for symmetric spaces.

Theorem : (Weinstein [1968][‡])

If M is a compact manifold without boundary, not homeomorphic to S^2 , then there exists a metric g on M and a point $p \in M$, such that the cut locus $\text{Cut}(p, g)$ has no conjugate point of p along any distance-minimizing geodesic from p .

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We have generalized this to the case of submanifolds.

Theorem : (Prasad, Schick[§])

Suppose M is a compact, connected, manifold without boundary, with $\dim M \geq 3$. Let $N \subset M$ be a closed, embedded submanifold, each component being without boundary, and having codimension ≥ 1 . Then, there exists a metric g on M such that the cut locus $\text{Cut}(N, g)$ contains no focal point of N along an N -segment.

[†] Harry Ernest Rauch. *Geodesics and curvature in differential geometry in the large*. (1959).

[‡] Alan D. Weinstein. *The cut locus and conjugate locus of a riemannian manifold*. Ann. of Math. (1968).

[§] Aritra Bhowmick, Sachchidanand Prasad and Thomas Schick. *Intersection of Cut and Focal Loci of a Submanifold*. (in preparation).

An Estimate on the First Focal Time

Principal curvature : generalization of the curvature of a planar curve.

Sectional curvature : generalization of the Gaussian curvature of a surface.

Theorem : (Warner [1966][†])

Let $H \subset M$ be a codimension 1 submanifold in a compact Riemannian manifold (M, g) , and $\mathbf{n}$ be a unit vector field normal to H at every point. Suppose $\Delta_H, \kappa_M > 0$ are real numbers such that

- the principal curvatures of H along $\mathbf{n}$ are $> -\Delta_H$, and
- all the sectional curvatures of M are $< \kappa_M$.

Set

$$\varepsilon_H := \frac{1}{\sqrt{\kappa_M}} \arctan \left(\frac{\sqrt{\kappa_M}}{\Delta_H} \right).$$

Then, for any unit speed geodesic $\gamma : [0, \infty) \rightarrow M$ in the direction of $\mathbf{n}$, there is no focal point of H along γ in the interval $[0, \varepsilon_H]$.

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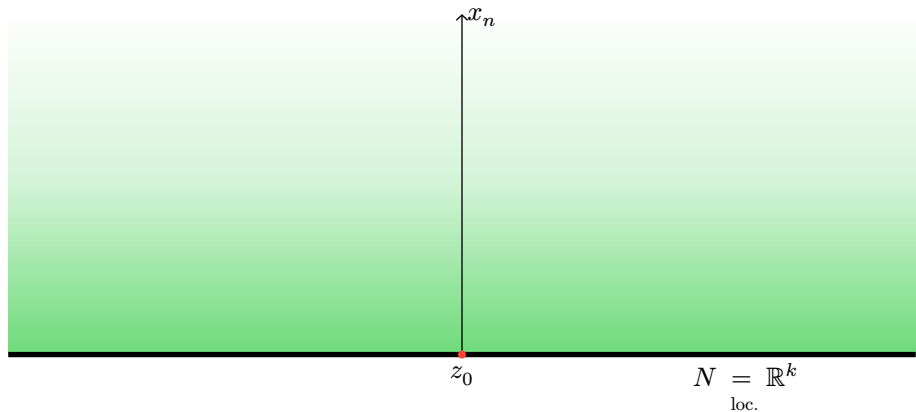
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Step 1: Choosing an Initial Metric and a Tubular Neighborhood

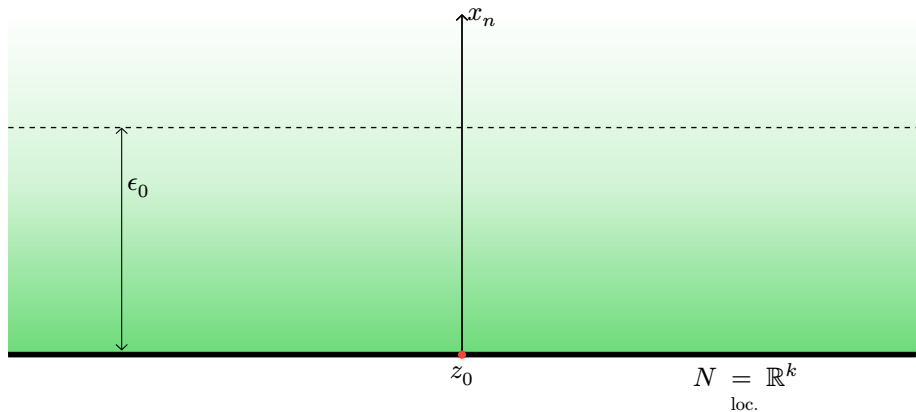
For $z_0 \in N$, get a metric which is Euclidean near z_0 and N is isometric to $\mathbb{R}^k \times \{0\} \subset \mathbb{R}^n$. Choose an ϵ_0 -tubular neighborhood of N , which is $\mathbb{R}^k \times \mathbb{D}_{\epsilon_0}^{n-k}$ near z_0 . Get a smaller tubular neighborhood Ω , whose boundary P is flat and parallel to $\mathbb{R}^{n-1} \times \{0\}$ near z_0 .



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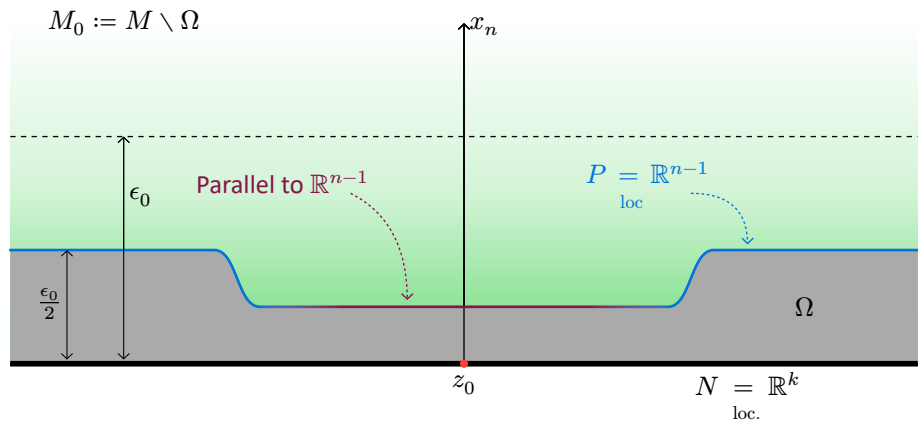
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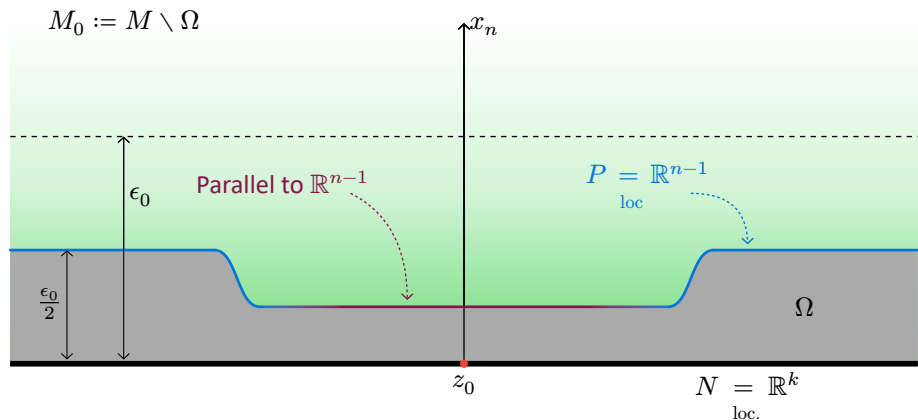
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Theorem : (, Prasad, Schick)

There exists $\Delta_{\text{curve}} > 0$, such that we can get a regular curve in $M_0 \setminus P$, with geodesic curvature strictly bounded by Δ_{curve} , which is arbitrarily dense in a given compact set $\mathcal{K} \subset M_0 \setminus P$. The curve can be assumed to be embedded if $\dim M_0 \geq 3$.

Fix $\Delta_P, \kappa_M > 0$, such that principal curvatures of P are $> -\Delta_P$ along the outward normal directions, and sectional curvatures of M are $< \kappa_M$. Set,

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Get ε_0 -neighborhood of P , and set $\mathcal{K}$ as the region away from N , with boundary Q .

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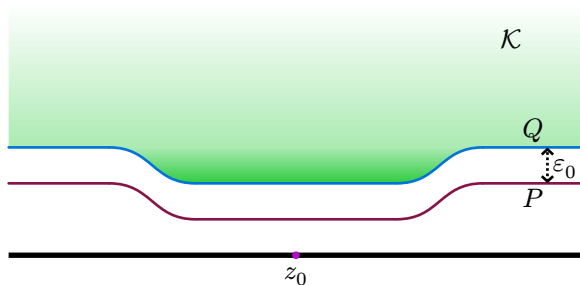
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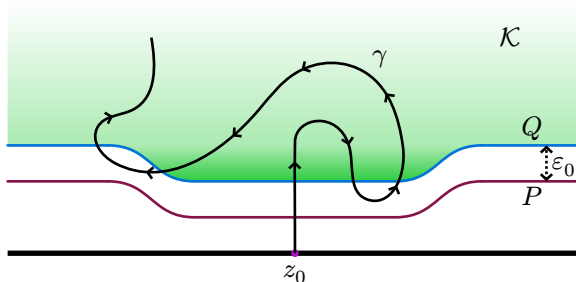
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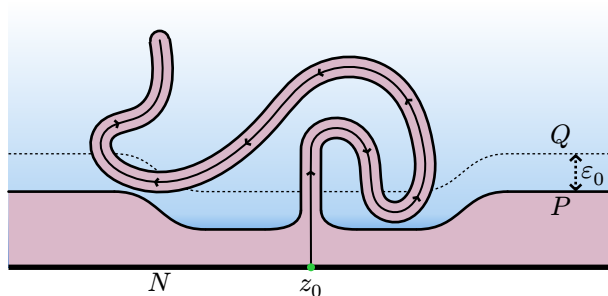
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The new region $\tilde{\Omega}$ is a tubular neighborhood of N , with boundary $\mathcal{B}$. We estimate the principal curvatures of $\mathcal{B}$.

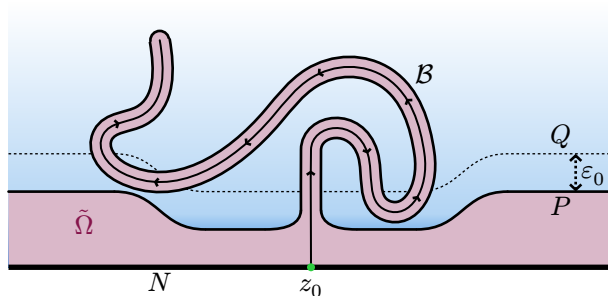
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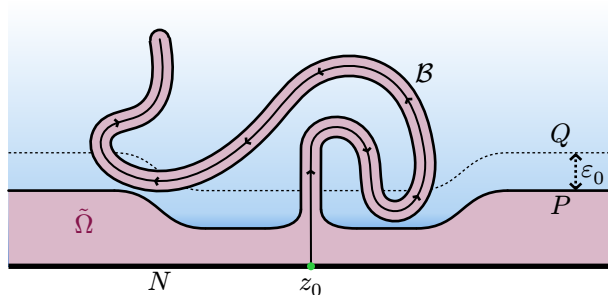
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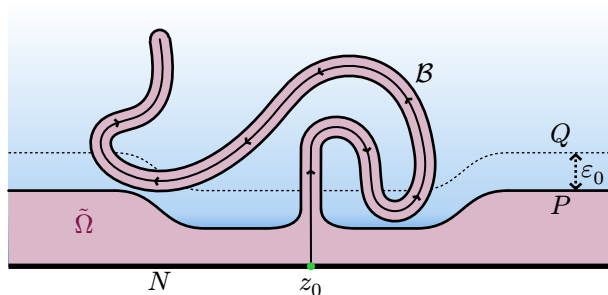
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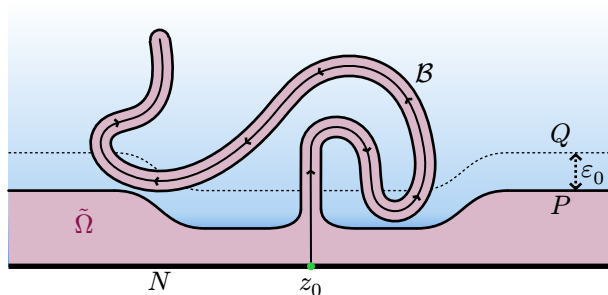
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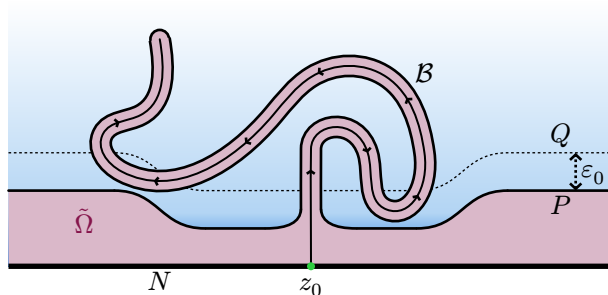
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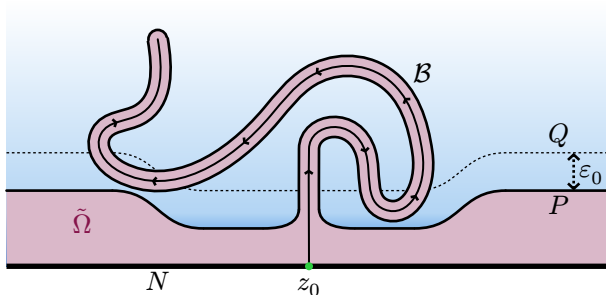
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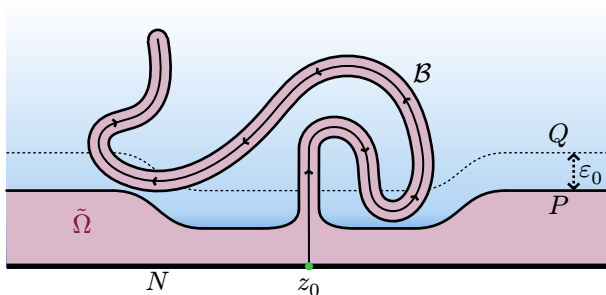
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Theorem : (, Prasad, Schick)

There exists a new metric g on M such that the following holds.

- g is unchanged on the complement of the tubular neighborhood $\tilde{\Omega}$, near its boundary $\mathcal{B}$, and near the submanifold N .
- $\tilde{\Omega}$ is a *unit normal neighborhood* of N with respect to g .

Proof of Main Theorem : With the new metric g ,

- every $\mathcal{B}$ -segment $\gamma : [0, \epsilon] \rightarrow M_0 \setminus \tilde{\Omega}$ extends back to an N -segment $\tilde{\gamma} : [-1, \epsilon] \rightarrow M$, and
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Consequently,

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Both of them can be generalized under the framework of **wind Finsler geometry**.

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Our immediate goals are following.

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Recent works of M. A. Javaloyes and his collaborators have shown the potential of modeling wildfire using wind Finsler geometry, where one can take into account both wind direction and slope of the ground.

Our immediate goals are following.

- Can we extend the joint work with S. Prasad[†] to the setup of wind Finsler geometry?
- Can the stability of the cut locus be proved in the context of Lorentzian geometry?
- Under the assumption that the cut locus is a finite subcomplex, can we bring some algebraic topology machinery into play?

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Ongoing Work

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Fibrations with a Section

Given a fibration $F \xrightarrow{\iota} E \xrightarrow{p} B$ with a (homotopy) section $s : B \rightarrow E$, there is a natural map

$$\begin{aligned}\theta : \Omega B \times \Omega F &\rightarrow \Omega E \\ (\gamma, \eta) &\mapsto (s \circ \gamma) \star (\iota \circ \eta),\end{aligned}$$

where ΩX denotes the loop space of a (based) space X . Classically, it is known that θ is a homotopy equivalence. The loop concatenation makes loop space into an H -space.

Theorem : (Basu, Samanta [2025][†])

θ induces a Lie algebra map in the homotopy groups (where the Lie brackets are given by Whitehead brackets) if and only if the James brace products $\{, \}_s : \pi_i B \times \pi_j F \rightarrow \pi_{i+j-1} F$ vanish identically.

θ is an H -map if and only if the *generalized* James brace product map vanishes identically.

In the paradigm of Stasheff, H -spaces are A_2 -spaces, and a loop space is actually an A_∞ -space. A natural question is :

when is θ an A_n -map, for $2 < n \leq \infty$?

Initial investigations indicate connections with polytopes known as resultohedra, their higher homotopy associative generalizations, and with the power and product maps of monoids.

[†] Somnath Basu, Aritra Bhowmick and Sandip Samanta. *On the James brace product: Generalization, relation to H -splitting of loop space fibrations & the J -homomorphism* Indiana Univ. Math. J. (2025).

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Courses that I can teach:

- Real and complex analysis
- Group theory
- Linear algebra
- Commutative algebra
- Topology
- Differential geometry
- Algebraic topology
- Homological algebra and category theory

Courses that I would like to develop:

- Introduction to Riemann-Finsler geometry
- Introduction to polytopes and homotopy theory
- Cut locus in computer graphics

Thank you!